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Virtual reality for gender microaggression training: A qualitative exploration of higher education students' immersive learning experiences

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ABSTRACT

Gender-based microaggressions remain a pervasive challenge for women, compromising their success in academia and the workplace. Despite increased awareness of overt sexism, these subtle forms of bias often go unaddressed, highlighting the need for innovative training approaches that equip individuals with the skills needed to counteract such behaviors. Virtual Reality (VR) has emerged as a promising medium for Diversity, Equity, and Inclusion (DEI) training, offering immersive simulations of real-world scenarios that allow learners to engage in complex social interactions within a computer-controlled environment. This study qualitatively explores higher education students' experiences with VR DEI training through a gender inclusion module focused on navigating microaggressions. Using semi-structured interviews and reflexive thematic analysis, four key themes were developed: (a) immersion and engagement afforded by the VR medium, (b) perceived realism and contextual fidelity of the scenario, (c) perceived learning outcomes, and (d) the practical and technical factors that influenced training delivery. These findings highlight the potential of VR as a pedagogical tool for DEI training, particularly its ability to provide experiential learning opportunities. However, these findings also reveal areas for improvement, including the need for dynamic and adaptive interactions, greater contextual fidelity, and personalized learning tailored to individual learner needs. By addressing these challenges, VR can serve as a practical and impactful training tool for fostering inclusive environments.

1. Introduction

Microaggressions are "subtle verbal, behavioral, or environmental slights that communicate derogatory messages and insults toward marginalized or under-represented groups" (Lilienfeld, 2017; Sue et al., 2007; Williams, 2020). They are often rooted in stereotypes and can appear as sexism, racism, or other forms of bias in daily life (Haynes-Baratz et al., 2021; Sue & Spanierman, 2020). Although often unintentional, microaggressions have a profound impact on the mental health and well-being of those who experience them, contributing to stress, burnout, and absenteeism (Ayari, 2023; Haynes-Baratz et al., 2021). Among these, gender microaggressions are particularly widespread, reinforcing traditional gender roles and undermining women's advancement in both academic and professional settings. Despite growing awareness of overt sexism and its negative effects (Kim & Roberson, 2022; Sue & Spanierman, 2020), subtle gender biases continue to persist. Women often report being interrupted, excluded from opportunities, or having their contributions overlooked unless echoed by male colleagues (Beltran et al., 2021; Haynes-Baratz et al., 2021), highlighting the persistent nature of gender microaggressions across contexts.

Because these microaggressions remain common and harmful throughout women's lives, addressing them is vital for creating inclusive environments where all individuals can thrive. DEI training offers one potential approach by promoting awareness, encouraging positive interaction, and reducing prejudice (Bezrukova et al., 2016). Traditional DEI programs typically use lectures, discussions, and workshops to introduce the concepts of bias and promote awareness (Gino & Coffman, 2021). While these methods increase understanding, they often fall short of opportunities for developing the interpersonal skills necessary to respond effectively to microaggressions in real-world situations. Research suggests that successful training should not only promote awareness but also provide opportunities to practice and develop the skills required for behavioral change (Bezrukova et al., 2016; Gino &

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Coffman, 2021; Kim & Roberson, 2022).

VR has emerged as a promising medium to address these gaps by offering an immersive and interactive learning environment (Makransky & Petersen, 2021) that enables learners to practice complex interpersonal skills that are often difficult or too sensitive to replicate through traditional methods (Dubiel et al., 2025). In the context of DEI, learners can engage in simulated real-world scenarios where interpersonal dynamics play a critical role, without the risk of real-world consequences. This form of experiential practice may facilitate the transfer of skills to real-life contexts where such interactions naturally occur (Grossman & Salas, 2011). Recent studies indicate that VR holds promise for fostering empathy and promoting inclusive behaviors across DEI-related contexts, including unconscious bias and microaggressions (Bursztein et al., 2024; Hamed et al., 2024), racism (Medvec et al., 2023), and sexual harassment (Rawski et al., 2022).

While an increasing body of research supports VR's potential for addressing sensitive social issues, an important gap remains in understanding how learners experience and make sense of DEI scenarios in immersive environments. Little is known about the factors that influence perceived realism and emotional engagement, or how learners distinguish between experiences that feel authentic and those that seem artificial within these environments. To address this gap, the present study explores the immersive learning experiences of higher education students preparing to enter the workforce, to provide rich, learner-centered insights into what works, what does not, and why.

This study uses a VR gender microaggression module that simulates workplace interactions by placing learners in the role of a bystander and prompting them to act. Given the limited prior research in this area, our study makes a twofold contribution. First, we examine how well-established immersive learning concepts, such as immersion, presence, and engagement, apply to a VR DEI context, contributing new insights to this under-researched domain. Second, we provide rich, learner-centric data that identifies key elements influencing students' learning experiences with VR microaggression training, highlighting areas for improvement and offering design considerations for future practice.

2. Research questions

The objective of this qualitative research is to explore higher education students' experiences with VR gender microaggression training and to identify key factors that influence these experiences, including insights into what works and what does not from their perspective. To achieve this, our study is guided by the following research questions.

RQ1: What are learners' experiences with VR microaggression training?
RQ2: What key factors, from learners' perspectives, influence their learning experiences during VR microaggression training?

3. Methods

3.1. Study design

This study employed a qualitative research design to explore higher education students' experiences with gender microaggression training in VR. The research followed the Standards for Reporting Qualitative Research (SRQR; O'Brien et al., 2014) to ensure transparent and systematic reporting of the research process. A qualitative approach was chosen for its capacity to provide rich, detailed insights into participants' lived experiences, aligning with the exploratory nature of the study and addressing the limited research on VR-based DEI training. As noted by Elliott & Timulak, 2005, qualitative methods enable flexibility and depth, making them particularly suitable for investigating complex or underexplored phenomena such as VR-based DEI.

The decision to adopt a qualitative design also reflected the complex nature of gender microaggression training, which involves the interplay of interpersonal skills such as communication and empathy, all of which are shaped by personal and situational factors. Pistrang and Barker (2012) highlight that qualitative inquiry is well suited to capturing such nuances and generating in-depth textual data that can inform evidence-based design recommendations. Furthermore, examining VR as a medium for DEI training extends these insights to related contexts, such as workforce and soft skills development. Through qualitative data, this study sought to generate detailed insights into how the unique affordances of VR, both established and emerging, shape learning experiences in similar training contexts.

3.2. Participants

This study included 28 higher education students (14 women and 14 men) recruited through purposive sampling. Participants met two inclusion criteria: (1) proficiency in English and (2) current enrollment in a higher education program. Ages ranged from 21 to 34 years ($M = 25$, $SD = 3.50$). Participants represented 17 nationalities and were enrolled in master's ($n = 20$) and PhD ($n = 8$) programs. Recruitment was conducted through posters displayed on university campuses, emails sent to the university community, and word of mouth. Participation in this study was voluntary, and participants did not receive compensation. A booking system allowed participants to schedule preferred training sessions via a shared calendar link. The resulting sample represented a range of cultural and educational backgrounds, providing varied perspectives on immersive DEI training. Table 1 provides an overview of participant demographics.

3.3. Human subject ethical approval

This study was approved by the Ethics Committee of the Université de Paris Cité (CER), with reference number N°IRB: 00012023-78. All participants provided written informed consent before taking part in the study, and their rights to privacy, confidentiality, and voluntary participation were ensured throughout the research process in compliance with European data protection regulations (GDPR).

3.4. Data collection methods

Each participant completed a short pre-training questionnaire to collect demographic information (e.g., age, gender, nationality, and educational level). Qualitative data were collected through semi-structured interviews conducted immediately after each VR training session. A total of 28 individual interviews were carried out, one *per* participant, and all were audio-recorded with participants' consent. Each interview lasted approximately 30 min and was conducted in person in English within a dedicated training room. The semi-structured format provided both consistency across interviews and flexibility to explore participants' individual experiences and perceptions. As highlighted by Kallio et al. (2016), this approach is particularly effective for addressing complex and, for some, emotionally sensitive topics, such as gender microaggression, while allowing diverse perspectives to be expressed.

Table 1
Participant demographics.

Characteristic	Value
Age	$M = 25$, $SD = 3.50$
Gender	Men: 14 (50%) Women: 14 (50%)
Education level	Master's: 20 (71.4%) PhD: 8 (28.6%)
Nationalities represented ($n = 17$)	American (6), British (1), Cameroonian (1), Chinese (1), Colombian (1), French (3), Ghanaian (1), Indian (4), Italian (1), Lebanese (1), Mexican (1), Pakistani (1), Russian (1), Spanish (1), Syrian (1), Thai (1), Turkish (1)

The interview guide comprised 22 questions (see Appendix A) covering key themes related to participants' experiences, including emotional responses during the training, perceived realism of the scenario, and encountered difficulties. This flexible format enabled the researchers to explore participants' perspectives in depth and to ask probing questions where clarification or elaboration was needed.

3.5. Training scenario and delivery

This study employed *Bodyswaps*' Gender Inclusion module, a VR training focused on workplace gender microaggressions (Bodyswaps, 2023). The training aimed to help learners recognize workplace microaggressions and practice constructive strategies to address them. The module consisted of three sequential VR scenarios forming a short narrative. In these scenarios, participants joined a virtual project meeting with two colleagues, Sam and Sophie. Across the scenarios, participants first observed and identified microaggressions, then reflected on them, and finally practiced giving feedback.

3.5.1. Microaggressions in the workplace

In the first scenario, participants played the role of a bystander during a project update meeting between Sam and Sophie. Their task was to observe and identify instances of microaggressions exhibited by Sam toward Sophie. Eight microaggressions were embedded in the dialogue, including behaviors such as minimizing, dismissing, excluding, and marginalizing. Using the VR controller, participants marked these behaviors when they spotted them. The scene depicted Sam interrupting Sophie, questioning her commitment because of remote work and time off for her child, referring to her informally as "love," and disregarding her concerns about delayed deliverables. The scenario concluded with Sophie reaffirming her commitment to the project timeline before leaving the meeting (see Fig. 1).

3.5.2. Encouraging others to reflect on discrimination

After Sophie left, participants interacted directly with Sam. They were prompted to choose among predefined dialogue options to address Sam's behavior. Each response represented a different feedback strategy. Selecting a suboptimal response triggered immediate feedback explaining why the choice was less effective, whereas selecting the optimal response produced feedback highlighting its appropriateness and potential to encourage reflection (see Fig. 2).

3.5.3. Providing feedback

In the final scenario, participants delivered verbal feedback directly to Sam. After tapping the "Record" button, they spoke their message aloud, and the system recorded it. The module then activated the *body swap* feature, enabling participants to view their own character delivering feedback to Sam. This perspective allowed them to experience tone, phrasing, and non-verbal communication from the recipient's point of view. Upon completion, the system provided a performance analysis covering aspects such as eye contact, tone, and overall delivery (see Fig. 3).

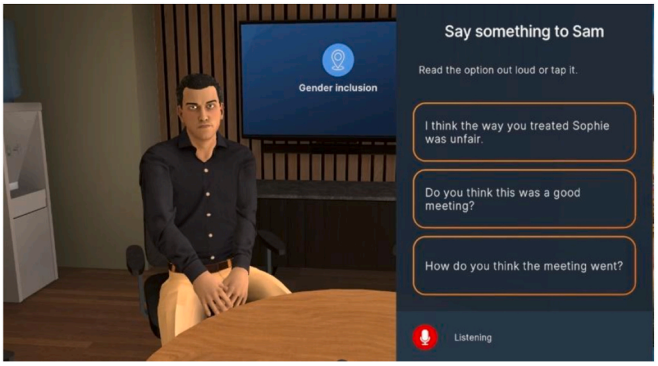


Fig. 2. Scene in which participants prompt Sam to reflect on his behavior through predefined dialogue options.

3.6. Procedure

At the beginning of each session, participants met the researchers in the training room at their scheduled time slot. Participants were welcomed and provided with an information sheet describing the study's purpose and procedures. They then received a consent form and a brief demographic questionnaire. After completing the questionnaire, participants were instructed to put on a Meta Quest 2 headset (see Fig. 4), which allowed them to become fully immersed in the virtual environment and experience the scenarios from a first-person perspective.

The researchers provided guidance and clarified that there was no right or wrong way to respond during the training; the aim was to explore participants' experiences. Participants were assisted in locating the gender inclusion training module within the application and initiating it by clicking the "Start" button. Once the training began, participants completed the session autonomously, while the researchers remained available to assist in case of technical difficulties or participant queries. The VR training lasted approximately 10–15 min. After completing the VR experience, participants removed their headsets and took part in a post-training semi-structured interview to provide detailed reflections on their learning experience. The overall session, including briefing, VR training, and interview, lasted approximately 45–60 min.



Fig. 1. Scene depicting the project update meeting between the virtual colleagues Sam and Sophie.

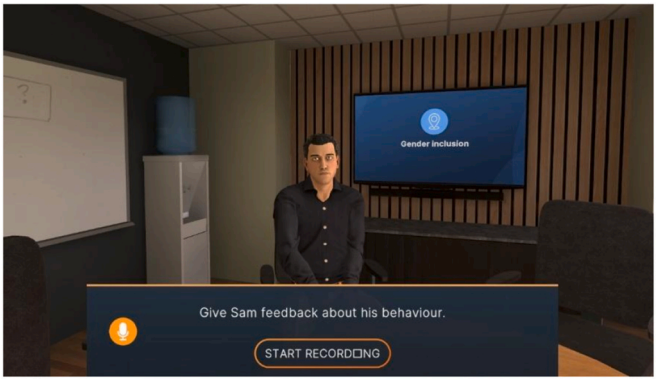


Fig. 3. Scene depicting the verbal feedback stage in which participants record spoken feedback directed to Sam.



Fig. 4. Participant wearing a Meta Quest 2 headset during the VR gender microaggression training session.

3.7. Data analysis

The audio recordings were transcribed verbatim by the primary author and a graduate research assistant. Throughout the process, participant confidentiality was maintained through the use of pseudonyms and secure data storage. Interviews were analyzed using reflexive thematic analysis (Braun & Clarke, 2019), with data management and coding facilitated by NVivo Qualitative Data Analysis Software (QSR International Pty Ltd., 2018). This approach recognizes the researcher's active role in the co-construction of meaning and treats reflexivity as central to analytic rigor. We followed the six-phase framework proposed by Braun and Clarke (2006) as an iterative and recursive guide, moving back and forth through the phases (Byrne, 2022). These phases included: (1) familiarizing ourselves with the data through active and repeated reading to gain an overall understanding, (2) generating initial codes by identifying meaningful patterns in the data, (3) generating themes by organizing codes across the dataset and combining them into overarching themes, (4) reviewing those themes to ensure they accurately represent the meanings in the data, (5) defining and naming the themes to capture the essence of each, and (6) producing the report by presenting the final themes and making an argument in relation to the research questions.

To support reflexive depth and coherence of the analysis, the primary author and a research assistant engaged in continuous analytic dialogue throughout data collection and analysis. After interviews, they held informal debriefing discussions to reflect on emerging impressions and potential meanings within participants' accounts. Both the primary author and research assistant independently engaged with the transcripts, producing interpretive notes that captured preliminary observations and meaning patterns in the data. The primary author maintained reflexive memos to document reflections on participants' experiences and record insights that enriched the analysis. An illustrative example showing how a data extract developed into a code and theme is presented in Appendix B.

4. Results

Through reflexive thematic analysis, four themes were developed that capture how participants engaged with and made sense of the VR training experience: (1) immersion and engagement, (2) perceived realism and contextual fidelity, (3) perceived learning outcomes, and (4) practical and technical considerations for training delivery (see Table 2).

4.1. Theme 1: immersion and engagement

Learners consistently described the VR training as an *immersive* and *involving* experience that made them feel as if they were "there" in the virtual environment. Those with prior experience in more conventional DEI training formats, such as online courses, video-based instruction, workshops, or lectures, frequently emphasized how the VR format compelled more active participation. As P1 explained, "This definitely feels more involved … you are more into the experience compared to other ones. VR forces you into a position where you have to say something and give feedback." Similarly, P27 noted, "As soon as you get into it, you are immersed. You play into it, rather than just watching a video."

Moreover, learners also valued that the VR environment enabled them to put their knowledge into practice. This allowed them to reflect on how they might respond in real-life situations by simulating the immediacy and spontaneity of real-world interactions. For example, P12 stated "… here you watch a conversation and have to respond to it with someone, and it's good to have immediate follow-up. In a theoretical situation, you have lots of time to carefully think about and perfect your response." Another participant (P13) highlighted this experiential learning opportunity, noting that it "provides both theoretical knowledge and a practical scenario of what you might encounter.''

At the same time, the immersive experience prompted a range of affective responses, particularly in the scenario where learners observed Sam's microaggressions toward Sophie. Some reported feelings of shock, annoyance, or discomfort. For example, one student (P9) reflected, "It was shocking for me to see him cutting her off like that." These reactions often translated into a strong dislike of the male character, Sam, whose behavior was described as "horrible" and "disrespectful." P1 remarked, "I had a strong dislike for that character," and P22 added, "Sam was so horrible, he kept saying very bad things to Sophie."

While participants reported the discomfort elicited by the scenario, the engagement with the VR medium itself was described as positive and

Table 2
Summary of themes developed through reflexive thematic analysis.

Theme	Definition/Focus	Illustrative Quote
1. Immersion and Engagement	Learners described experiencing immersion and greater engagement compared with traditional DEI formats. VR compelled active participation and evoked affective responses ranging from enjoyment with the medium and various emotions related to the training content.	*"This definitely feels more involved … you are more into the experience compared to other ones. VR forces you into a position where you have to say something and give feedback." (P1)*
2. Perceived Realism and Contextual Fidelity	Learners perceived differences in how realistic the VR scenario felt compared with real-life situations. While many found the technical environment realistic and immersive, others described the scenario itself as exaggerated or limited by missing contextual information and restricted user agency.	*"I believe it was an exaggeration … with a lot of microaggressions condensed into a single dialogue. I don't think it reflects what people experience in everyday life." (P6)*
3. Perceived Learning Outcomes	Learners reported increased awareness of microaggressions, understanding alternative response strategies, and reflection on their ability to communicate feedback effectively.	*"The training showed how to lead a conversation without attacking the person but asking questions that make them figure out what they did wrong." (P8)*
4. Practical and Technical Considerations for Training Delivery	Learners emphasized the need for repeated short sessions and varied scenarios. They also discussed usability challenges such as headset weight and visual clarity.	*"This is something you need to learn consistently, not just with a 10-min training, but rather 10 min per day." (P16)*

enjoyable. Learners highlighted that the interactive format made the training "fun," "pleasant," and emotionally engaging. For instance, P7 commented, "Clicking to spot microaggressions was really fun," while P4 shared, "It was like I was there in person, that was enjoyable."

Yet, for some participants, particularly those less familiar with VR, the novelty of the environment introduced challenges to maintain focus. A few described being distracted by the urge to look around and explore the virtual setting rather than fully attend to the dialogue. P12 explained this, "Because it was my first time using VR, it was hard to pay attention to what they were saying, as I just wanted to look around." Similarly, P21 described this as, "I was like, 'Wow, such a detailed scene around me,' but I think with more experience, I would really be able to focus on the learning process."

4.2. Theme 2: perceived realism and contextual fidelity

Learners frequently discussed how closely the VR scenario reflected real-life situations. Their comments focused not only on the technical or graphical aspects of the system, but also on the extent to which the computer-based context represented the social dynamics of real-world interactions and matched their expectations of a realistic scneario.

Several participants reported that the scenario depicting microaggressions from Sam toward Sophie did not feel fully realistic. They noted that, in their experience, microaggressions are typically more subtle and occur gradually rather than being concentrated within a single dialogue. For example, P6 explained, "I believe it was an exaggeration of certain situations, with many microaggressions condensed into a single dialogue. I don't think it reflects what people experience in everyday life." Similarly, P8 commented, "It was a bit unrealistic because microaggressions occur in very shady ways."

Another factor influencing participants' perceptions of realism, and how they believed they would respond in comparable real-life situations, was the amount of contextual information provided. Some emphasized the need for more background information about Sam, such as his identity and personal circumstances, to enable more realistic and meaningful interaction. Others highlighted the importance of understanding workplace power dynamics and their position relative to Sam, noting that their responses in real-life scenarios would differ depending on hierarchical relationships. Insufficient contextual details made it difficult for them to respond in ways consistent with how they would act in real situations. For example, P4 remarked, "I don't know Sam, I haven't worked with him or know anything about his children. Having more personal context about who he is would have allowed me to be more authentic and empathetic." Likewise, P18 stated, "In real life, my response to Sam would differ depending on whether I'm his colleague or his boss."

Participants also discussed how the level of agency afforded by the system influenced their experience of the scenario. Several expressed a desire to intervene during the microaggressions they witnessed, but the fixed scenario narrative limited their ability to respond as they would in real-life situations. As P2 explained, "I think in a realistic scenario, someone would point it out immediately instead of waiting until the end of the conversation." Similarly, P20 remarked, "They didn't allow me to say anything; it was just the two of them speaking. Since it's a meeting for all of us, I should be able to speak."

Moreover, while the predefined dialogue options aimed to demonstrate appropriate ways to address the situation, some participants felt constrained by the limited freedom to express themselves as they normally would. They explained that the predefined options did not fully align with how they would speak in a comparable real-world context. As P7 observed, "In real life, you have unlimited opportunities to speak and respond to the people you're conversing with. It also depends on how people react to you." Another participant (P18) emphasized the need for flexibility in social interaction, stating, "I do believe there should have been several correct answers, as in real life, situations like these don't have black-and-white answers."

Participants also discussed how these perceived limitations might affect the transfer of learning to real-life contexts. Many referred to these factors such as workplace hierarchies, personal relationships, and the unpredictability of human responses as key elements that would complicate applying the training in everyday contexts. For example, P1 explained this by stating, "My approach would depend on the person's history and their position, such as whether they are a superior or a peer. I believe there's a lot of nuances involved." Similarly, P20 added, "If I don't have any personal connections, I will call them out immediately. However, if it involves someone I have a relationship with, my approach would be gentler, guiding them toward what's right." Others noted that real-life interactions are often less predictable than the training scenario. They pointed out that personal characteristics, particularly those of individuals committing microaggressions, could significantly influence how participants would respond in real-life situations. Unlike the training scenario, where responses appeared more straightforward, real-world interactions are often more complex and unpredictable, especially when individuals are not receptive to feedback. As P2 remarked, "It depends on the person, specifically whether they are open to feedback or not," and P5 added, "In real life, asking the right question may not immediately change someone's attitude; it's not as straightforward as in the scenario."

4.3. Theme 3: perceived learning outcomes

Learners discussed their perceptions of what they had gained from the VR microaggression training. Their accounts reflected a range of perceived effects, including increased awareness of microaggressions, exposure to alternative response strategies, and opportunities for self-reflection on their communication and feedback.

Several participants described a perceived increase in awareness and recognition of microaggressions. In particular, those who were unaware of microaggressions or unsure of their nature noted that the training helped them identify behaviors that they might previously have overlooked. For example, P10 stated, "Before the training, I just did not like when people behave that way, but after the training, I could distinguish that this type of behavior is a microaggression." Similarly, P23 commented, "This has helped me to spot situations where we sometimes think 'no, this isn't a microaggression,' when in fact it is."

Participants who already had prior awareness of microaggressions described the training as introducing alternative, non-confrontational approaches to addressing them. They particularly referred to the second scenario, which demonstrated techniques for prompting reflection rather than directly confronting the male character. As P18 explained, "When I asked Sam questions, I was a bit aggressive in the beginning, but the training showed me how to not be aggressive." Likewise, P8 observed, "The training showed how to lead a conversation without attacking the person but asking questions that make them figure out what they did wrong."

However, while several participants valued this approach, others expressed a preference for more direct or assertive responses. Their willingness to adopt either approach depended on personal communication style and prior experiences. P20 described the training's approach as "very non-confrontational, like making a seven-year-old figure out why they should not eat too much candy … I prefer a more radical approach where I just confront what they have done." Similarly, P5 commented, "I would prefer continuing the direct way because I don't think they deserve such nice psychological treatment."

Some participants also mentioned that the training prompted reflection on how effectively they were able to communicate feedback. They described the final scenario, where they were asked to give feedback to Sam in their own words, as an opportunity to assess their comfort and confidence in this area. P5 reflected, "I realized that I am not good enough to give sufficient feedback," while P2 stated, "I realized I was not able to speak suddenly when I was asked to give feedback. I thought I was good at this, but I'm clearly not."

4.4. Theme 4: practical and technical considerations for training delivery

Participants discussed several practical and technical factors that influenced their experience with the VR training. Their comments primarily focused on two areas: the structure and delivery of the training, and the usability of the VR headset.

Several participants emphasized the need for repeated sessions rather than single exposure to effectively transfer learning to real-life contexts. They valued shorter but more frequent exposures, noting that ongoing practice would better support skill development in identifying and addressing microaggressions. As one participant explained, "This is something you need to learn consistently, not just with a 10-min training, but rather 10 min every day" (P16). Another added that "several training sessions would prepare me better to apply what I've learned in real life, but a single 10-min session is not enough" (P18). Participants also expressed the importance of including multiple scenarios to reflect the variety of contexts in which microaggressions occur, which they believed would strengthen their confidence in applying these skills.

In addition to delivery, participants highlighted technical and physical challenges related to the VR headset. Several described discomfort due to the headset's weight, noting it was acceptable for short sessions but potentially problematic during longer use. As one participant mentioned, "The headset is quite heavy. After a few minutes, I want to take it off" (P12). Others mentioned minor visual issues, such as blurriness or watery eyes, which occasionally disrupted their sense of immersion. These comments highlight the importance of considering usability factors when designing immersive learning experiences.

5. Discussion

This study explored how higher education students experienced a gender microaggression training in virtual reality. Through reflexive thematic analysis, four themes were developed that capture how participants engaged with and made sense of the immersive learning experience: (1) immersion and engagement afforded by the VR medium, (2) perceived realism and contextual fidelity of the scenario, (3) perceived learning outcomes, and (4) the practical and technical factors that influenced training delivery.

Participants often described the VR training as an engaging and immersive experience. The technology's immersive qualities evoked a sense of presence, with several learners noting that they felt "there" in the virtual meeting, as if seated alongside their virtual colleagues. According to the Cognitive Affective Model of Immersive Learning (CAMIL; Makransky & Petersen, 2021), presence, or the feeling of "being there", is one of the main affordances of immersive environments that can support both cognitive and affective engagement. Consistent with this model, feeling present in the virtual environment seemed to encourage learners to participate more actively and engage on an emotional level, unlike the more passive experiences they associated with traditional DEI sessions such as lectures or pre-recorded videos.

To better understand the specific nature of the presence experienced by participants, Slater's (2009) distinction between place illusion and plausibility illusion offers a useful lens. The sense of "being there" corresponds to place illusion, the feeling of being physically located in a virtual space. While this form of presence was frequently reported, many participants described disruptions to plausibility illusion, which concerns whether events in the virtual setting are perceived as genuinely occurring, even when users know they are computer-generated (Rovira et al., 2009; Slater, 2009). In this sense, plausibility is not dependent on physical fidelity but on the perceived credibility, relevance, and social coherence of the scenario. Although a strong sense of place illusion may allow users to feel present, plausibility plays a key role in determining whether users respond to the situation as if it were real. When plausibility was interrupted, participants described a noticeable "reality gap" (Rovira et al., 2009) that likely influenced their engagement and the authenticity of their responses.

Several factors appeared to contribute to these breaks in plausibility. The most common concern was scenario design. Many participants perceived the microaggressions as overly explicit or condensed, lacking the ambiguity that often characterizes real interpersonal encounters. As a result, they found it easier to recognize problematic behavior but more difficult to connect the situations to their lived experiences. Participants also remarked on insufficient contextual details such as workplace hierarchies or background information about the characters, which made it harder to interpret the social dynamics within the scene. This insufficient contextual information appeared to limit their ability to respond naturally. Moreover, the fixed and linear narrative design seemed to restrict learner agency, as they were unable to act or intervene as they would prefer in real-life contexts. Within CAMIL, agency is defined as the feeling of generating and controlling one's actions, representing another key affordance of immersive learning environments (Makransky & Petersen, 2021). Participants' frustration at being unable to intervene as they saw fit suggests that a lack of agency can disrupt both the plausibility of the experience and the potential for authentic practice. This aligns with prior work highlighting the complex relationships between interactivity, learners' sense of agency, and learning-related experiences in VR environments (McGivney et al., 2025).

These experiential factors, particularly the scenario's plausibility and the degree of agency it afforded, appeared to shape participants' perceived learning outcomes, which were also influenced by their prior knowledge and individual communication styles. For many, the scenario's lack of subtlety, while reducing plausibility for some, still provided a useful starting point for recognizing microaggressive behaviors, especially among those with less exposure to DEI concepts. In contrast, participants who were already familiar with the topic described the main value of the training as the opportunity to consider alternative, non-confrontational ways of addressing such situations. This approach, however, did not resonate with everyone, as the limited agency within the simulation seemed to prevent some participants from practicing the more direct strategies that felt authentic to their usual communication styles. These varied accounts suggest that participants evaluated the training's value through the interplay between the simulation's design features and their own experiences. In this sense, the training's value appeared to lie not in prescribing a single correct response but in providing a simulated space for learners to reflect on, practice, and explore the strategies that felt most authentic and effective to them.

5.1. Design implications for immersive DEI learning

Our findings suggest that VR can be a valuable medium for learning that involves complex social interactions, particularly in DEI training where real-world practice can be challenging. VR allows learners to engage in experiential learning through interactive role-play within social simulations that promote reflection and skill rehearsal without real-world risk and consequences. The findings highlight that meaningful learning in VR depends not only on immersion but also on how instructional design creates authentic and contextually rich opportunities for practice. The qualitative insights from this study identify several areas for design consideration that could enhance the educational potential of immersive DEI training.

5.1.1. Enhance contextual fidelity to support plausibility

Our findings suggest that contextual fidelity plays an important role in influencing the plausibility of immersive DEI training scenarios. Participants noted that insufficient details about workplace hierarchies, character relationships, or organizational norms made interactions feel artificial and difficult to interpret or relate to their real-world experiences. This suggests that designers may benefit from integrating relevant social and contextual cues such as explicit roles, relationships, or prior events to anchor the scenario in recognizable social dynamics. Fidelity in this context is not about perfect graphical realism but about providing

enough background for the simulation to feel plausible and reflect the social realities learners recognize. Plausibility is essential for enabling authentic practice in VR-based DEI training. However, contextual richness should be balanced with pedagogical clarity, as excessive detail may increase cognitive load (Sweller, 2011).

5.1.2. Foster learner agency through flexible narrative design

We found that the scenario's fixed and linear structure may limit agency, influencing how participants experience their ability to act. Several participants described wanting to intervene earlier but were constrained by the system, which controlled when and how they could respond. Learner agency in immersive DEI training appears to depend less on the number of available options and more on being able to act intentionally at moments they consider appropriate. Designers might therefore consider more flexible narrative structures, such as branching pathways, which enable timely and contextually appropriate actions. This approach allows learners to explore different strategies, such as speaking up immediately, remaining silent, or addressing an issue privately, and to reflect on the consequences of each choice. In this context, agency functions not only as an interactional feature but also as a pedagogical condition that supports practice in navigating complex and unpredictable social situations.

5.1.3. Personalize and adapt the training content

Our findings show that participants' engagement was influenced by their prior knowledge, communication styles, and personal experiences. A single, uniform scenario did not resonate equally with everyone. Clear and explicit examples were helpful for learners new to DEI concepts, while those with more experience sometimes found them overly simplified or felt that the prescribed non-confrontational responses did not fully align with their preferred communication styles. These differences suggest that designers could consider adopting a more adaptive approach that personalizes training content. This could involve adjusting the complexity of scenarios, for instance by including both subtle and direct examples of bias, or offering alternative pathways that reflect different levels of familiarity and confidence. The goal is to maintain relevance and engagement for a broader range of learners, ensuring the training feels meaningful and applicable to their professional contexts and prior experiences rather than offering a single, uniform experience.

5.1.4. Explore generative AI for dynamic social simulations

Our findings revealed that the predefined dialogue options influenced how participants experienced the interaction, particularly in relation to practising authentic social exchanges. Some participants felt that the fixed responses were overly structured and did not fully to capture the unpredictable and context-dependent nature of real conversations. Because DEI skill-building relies on navigating complex interpersonal communication (Richter et al., 2023), the quality of interaction becomes a crucial area for design. These insights suggest that designers could explore more adaptive forms of dialogue interaction in immersive training environments. One potential approach is the use of generative AI, particularly large language models (LLMs), to support more adaptive and dynamic social simulations. Recent developments show that LLMs can generate contextualized responses to learners' input, supporting more natural, real-time interactions (Özkaya et al., 2025; Fink et al., 2024) that some participants in this study found lacking. However, this potential must be approached with ethical and pedagogical care. Research on generative AI also highlights the need for clearer guidance and policies regarding its use in learning environments (Smith et al., 2024).In DEI contexts, dialogue generation must therefore be carefully managed to ensure that AI-produced language remains accurate, respectful, and free from bias.

5.1.5. Improve practical delivery and usability

Our findings on practical delivery and usability highlight several implementation considerations that influenced participants' experiences. Many participants felt that a single short session was insufficient for developing practical skills, suggesting that training may benefit from being delivered through shorter, repeated sessions. This approach may help accommodate the physical limits of the hardware and allows skills to develop gradually. Participants also emphasized the need for varied scenarios to build confidence in different situations.

Technical issues also shaped the learning experience. Consistent with earlier studies (Huang et al., 2021; Miguel-Alonso et al., 2024), first-time users were often distracted by the novelty of the medium, spending time exploring the environment instead of focusing on the dialogue. Including short orientation exercises before training can help users become familiar with the technology and reduce these distractions. Finally, basic usability issues such as headset weight and visual clarity were also mentioned by participants. Ensuring that the equipment is comfortable, accessible, and easy to use is therefore important for supporting an effective learning experience.

6. Limitations

While this study provides valuable insights into how higher education students experience gender microaggression training in virtual reality, it is not without limitations. First, the study employed a single short VR session lasting approximately 10–15 min. Although this allowed for an in-depth exploration of participants' immediate impressions, it limits generalizability and prevents conclusions about long-term learning or behavior transfer. A single exposure is unlikely to produce sustained attitudinal or behavioral change, and the findings should therefore be interpreted as reflecting participants' perceived rather than enduring learning outcomes. Second, as in many interview-based studies in sensitive topics, social desirability bias (Nederhof, 1985) may have influenced participants' responses. This bias refers to the tendency to deny socially undesirable traits or reactions (for instance, not feeling disturbed by the microaggressions) and to report attitudes or emotions perceived as socially acceptable. Consequently, some learners may have emphasized socially acceptable reactions, such as strong disapproval of the male character's behavior. This possibility should therefore be considered when interpreting participants' accounts. Third, the participant group consisted of 28 higher education students. Although the sample was culturally diverse, it does not represent the broader workforce population. Participants' educational level and familiarity with DEI concepts may have influenced how they interpreted and engaged with the training scenarios. Therefore, the findings may not fully generalize to other age groups, educational background, or professional environments. Finally, the study adopted a qualitative, exploratory design focusing on participants' perceptions. While this approach offered rich, detailed accounts of their experience, it did not assess knowledge gain or behavioral change. As a result, conclusions about effectiveness must be limited to perceived rather than demonstrated outcomes. Future research could combine qualitative and quantitative approaches to examine whether participants' perceived learning outcomes align with actual performance or behavioral change.

7. Conclusion

The results of this study provided valuable insights into learners' experiences with VR microaggression training and identified critical factors to guide future design and research practices. The findings underscored the unique characteristics of VR, such as immersion, presence, and engagement, which can contribute to creating a safe and controlled environment where real-life scenarios can be simulated and practiced. However, the study also highlighted the need for more thoughtful and pedagogically sound approaches to designing the events and interactions within these virtual environments.

In this study, we explored learners' experiences with VR training for addressing gender microaggressions in workplace scenarios. Through qualitative research using semi-structured interviews, we identified four

key themes that offer a detailed understanding of the factors influencing their experiences in immersive DEI contexts. These findings highlight areas for improvement, such as enhancing the realism and contextual fidelity of scenarios, integrating dynamic and adaptive interactions, and personalizing training experiences to better meet the diverse needs of learners. Overall, this research underscores the potential of VR to support DEI training by offering experiential learning opportunities that traditional methods often cannot provide. By addressing the identified challenges and implementing the recommended strategies, future research and practice may better leverage the potential of VR as a tool for promoting inclusion and equity in educational and workplace settings.

CRediT authorship contribution statement

Esra Çakı: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Franck Zenasni:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Huyen Nguyen:** Writing – review & editing, Supervision, Resources, Conceptualization.

Statements on open data and ethics

This research was approved by the Ethics Committee of the Université de Paris Cité (IRB No. 00012023-78). All study procedures followed institutional policies, GDPR regulations, and established ethical standards for research involving human participants. Participation was entirely voluntary, and informed consent was obtained from all participants before data collection. To safeguard confidentiality, all materials were anonymized and stored securely in a restricted-access institutional repository. The study was conducted in full compliance with ethical principles and the legal requirements of France and the European Union.

Declaration of competing interest

The authors declare no competing interests related to the research study titled 'Virtual Reality for Gender Microaggression Training: A Qualitative Exploration of Higher Education Students' Immersive Learning Experiences', including authorship and the publication of this article.

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Appendix A

Interview guide used for the semi-structured participant interviews.

Semi-structured Interview Questions
1. Have you had any prior experience using virtual reality technology or headsets?
2. How did you feel during this virtual reality experience?
3. Were there any particular difficulties or obstacles you encountered while using the virtual reality technology during the training?
4. Have you ever participated in any kind of gender inclusion or diversity training or VR training before? If yes, what kind of training?
5. How does this VR training compare to previous training you've done?
6. What is your overall impression and thoughts about the training?
7. What aspects of the training program did you find the most helpful or effective?
8. Were there any aspects you found challenging or confusing?
9. Do you think the scenarios presented in the training were realistic and relevant to real-world situations?
10. What aspects of the virtual reality experience did you find most interesting?
11. Do you feel that the skills and knowledge you gained from the training would be transferable to real-life situations? If so, why?
12. What were your feelings and thoughts about the particular situations presented in the training?
13. Have you ever encountered a situation similar to this? How did you react?
14. How would you handle a similar situation in the future?
15. Were there any moments you felt uncertain or unsure about what to say to Sam?
16. Can you explain in detail how you approached and guided the conversation with Sam?
17. How did you feel at the moment you were talking to Sam? Did you feel confident addressing his behavior directly?
18. Do you think your response to the situation would have been different if it had happened in real life? How and why?
19. How can you see yourself using the skills you've learned in this training in your everyday life?
20. Do you feel like the training adequately prepared you to apply these skills in real-life situations?
21. What potential challenges might prevent you from applying the skills you've learned in real-life situations?
22. Are there any particular strategies or techniques you learned in the training that you think will be challenging to apply in real-life situations?

Appendix B

Illustrative Example of Reflexive Thematic Analysis Process

The following examples show how raw interview data were interpreted through reflexive thematic analysis (Braun & Clarke, 2019). Each excerpt is presented with the initial codes and its contribution to the final theme. The reflexive notes describe early analytic reflections rather than fixed interpretations.

Illustrative examples of how interview data extracts were coded and interpreted during the reflexive thematic analysis process.

Example 1: Immersion and Engagement			
Data Extract	Generated Code(s)	Reflexive Note	Contribution to Theme
P01: *"It definitely feels a bit more … you're more into the experience than in other ones I've done. … Doing it in VR was good because it forces you into a position where you have to say something and give feedback. Whereas it's easier, like if you're in your cubicle on your laptop answering the questions, to say 'Oh, I'd definitely say something here.' But actually, I wasn't expecting them to say, 'Okay, record your feedback now,' and it's harder to come up with what you want to say."*	• Engagement in the training experience • Active participation • Performance pressure • Comparison with conventional DEI training	This extract is notable for how engagement is described as involving real-time response rather than visual immersion alone. The phrasing "forces you" points to an immediacy of interaction that also seemed to influence their experience of involvement within the scenario.	**This extract illustrates** how the VR training prompted active participation and spontaneous verbal responses, **in contrast to** the reflective but less demanding format of traditional online DEI modules. The immediacy of the scenario **appears to contribute** to a sense of being "in" the situation rather than passively observing it.
Example 2: Perceived Realism and Contextual Fidelity			
Data Extract	Generated Code(s)	Reflexive Note	Contribution to Theme
P02: *"It definitely felt he was going a bit overboard. … His behavior was not that micro. In a realistic scenario, somebody would immediately point it out instead of waiting for the whole conversation to end. … His behavior warranted somebody immediately pointing it out." say something here.' But actually, I wasn't expecting them to say, 'Okay, record your feedback now,' and it's harder to come up with what you want to say."*	• Overt rather than micro behavior • Exaggeration of scenario • Expectation of real-time intervention	This account draws attention to how scenario realism was associated with the subtlety of social dynamics **rather than environmental detail**. It suggests that plausibility, for some participants, was linked to how accurately **interpersonal exchanges mirrored real-life interactions**.	**Participants** questioned how realistic the scenario felt, pointing out that real microaggressions are often less direct and occur in more subtle ways. They also noted that real-life social situations would **allow individuals to intervene immediately**, which **they felt was somewhat constrained in the simulation**.
Example 3: Perceived Learning Outcomes			
Data Extract	Generated Code(s)	Reflexive Note	Contribution to Theme
P18: *"I didn't have any experience … and no, I never participated in diversity training before. … Some microaggressions were highly aggressive, and sexism I believe is worse when it's not that aggressive … Sometimes sexism can be hidden between [things]. So I don't believe this training session did touch this. … But the questioning Sam part, I really liked it. When I had asked Sam questions … it showed me how to not be aggressive. … It showed me that asking questions that make the other person discover himself is more effective than just saying 'you're wrong.'"*	• No prior experience with DEI training • Distinction between overt and hidden sexism • Learning through guided dialogue • Reflecting confrontational and constructive communication	This extract is useful for understanding how the training allowed the participant to recognize alternative and more constructive ways of responding to inappropriate behavior.	This extract illustrates a participant identifying a new, practical communication strategy. It points to a specific learning moment in which the participant recognized that asking reflective questions **may be more effective than direct criticism**.

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